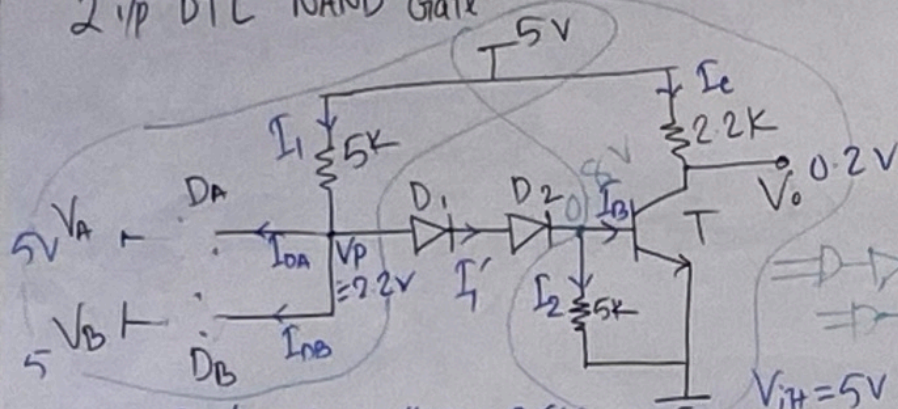


DTL (Diode Transistor Logic)

2 i/p DTL NAND Gate



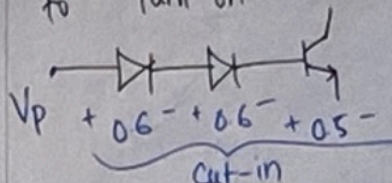
Diode cut-in voltage = 0.6V
 BJT cut-in voltage = 0.5V, $\beta_F = 30$, $V_{IL} = 0.2V$

- 1) Verify that the above ckt represents a NAND gate.
 Find V_o and all the currents for all input cases. β_{min}
- 2) Find Power consumptions, P_{avg} .
- 3) Find Maximum Fanout

$$P_4 = (5-0)(I_2 + I_3 + I_4) = 13.7 \text{ mW}$$

Ans

Necessary condition for transistor, T to turn on



if $V_p > (0.6 + 0.6 + 0.5) = 1.7V \rightarrow T \rightarrow \text{on}$
 if $V_p < 1.7V \rightarrow T \rightarrow \text{cutoff}$

V_A	V_B	A	B	V_o	Y
0.2	0.2	0	0	5V	1
0.2	5	0	1	5V	1
5	0.2	1	0	5V	1
5	5	1	1	0.2	0

$$P_{avg} = \frac{P_1 + P_2 + P_3 + P_4}{4}$$

Case-I

$$V_A = 0.2V, V_B = 0.2V$$

$D_A, D_B \rightarrow FB$

$$0.2V \xrightarrow{0.7V} V_P = (0.7 + 0.2) = 0.9V < 1.7V$$

$T \rightarrow \text{cutoff}$

$$I_1' = I_2 = I_B = I_C = 0$$

$$I_C = \frac{5 - V_o}{2.2k}$$

$$0 = 5 - V_o \rightarrow V_o = 5V$$

$$I_1 = \frac{5 - 0.9}{5k} = 0.82 \text{ mA}$$

$$I_{DA} = I_{DB} = \frac{I_1}{2} = 0.41 \text{ mA}$$

$$P_1 = (5 - 0.2) I_{DA} + (5 - 0.2) I_{DB}$$

$$= (5 - 0.2) \times I_1$$

$$= 3.936 \text{ mW}$$

Case-II

$$V_A = 0.2V, V_B = 5V$$

$D_A \rightarrow FB, D_B \rightarrow RB$

$$V_P = 0.7 + 0.2 = 0.9V < 1.7V$$

$T \rightarrow \text{cutoff}$

$$I_1' = I_2 = I_B = I_C = 0$$

$$I_C = \frac{5 - V_o}{2.2k} \Rightarrow V_o = 5V$$

$$I_1 = 0.82 \text{ mA}$$

$$I_{DA} = I_1 = 0.82 \text{ mA}$$

$$I_{DB} = 0 \text{ mA}$$

$$P_2 = (5 - 0.2) I_{DA}$$

$$= 3.936 \text{ mW}$$

Case-III

Same as II

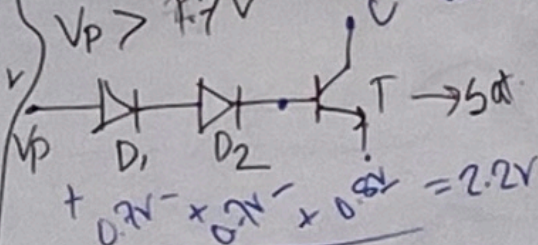
Case-IV

$$V_A = 5V, V_B = 5V$$

$$V_{BE} = 0.8V$$

$$V_{CE} = 0.2V$$

$$V_P > 1.7V$$



$$V_o = 0.2V \rightarrow V_{CE} = 0.2$$

$$I_{DA} = I_{DB} = 0 \text{ mA}$$

$$I_1 = I_1' = \frac{5 - 0.2}{5k} = 0.56 \text{ mA}$$

$$I_2 = \frac{0.8 - 0}{5k} = 0.16 \text{ mA}$$

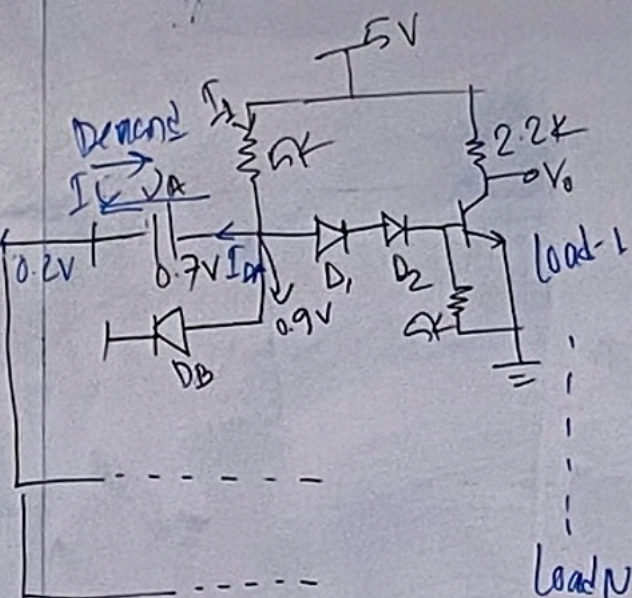
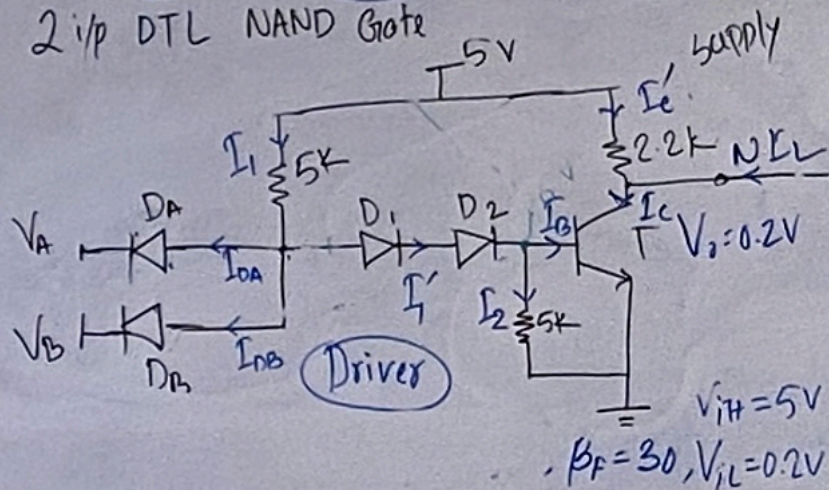
$$I_B = I_1' - I_2 = 0.4 \text{ mA}$$

$$I_C = \frac{5 - 0.2}{2.2k} = 2.18 \text{ mA}$$

$$\beta_{min} = \frac{I_C}{I_B} = 5.45$$

DTL (Diode Transistor Logic)

2 i/p DTL NAND Gate



- 1) Verify that the above ckt represents a NAND gate.
Find V_o and all the currents for all input cases. β_{min}
- 2) Find Power consumptions, P_{avg} .
- 3) Find Maximum Fanout

$N \rightarrow \text{Fanout}$

* Fanout

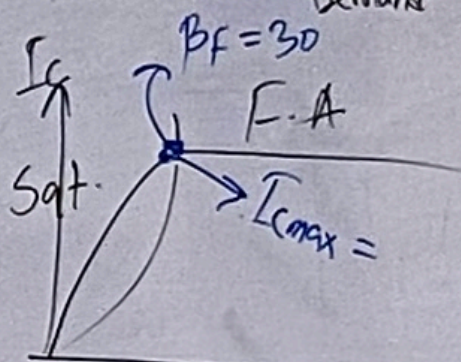
Case-I

$$V_o = \text{high} = 5V$$

$$\text{Supply current} = 0 = \frac{5-5}{2.2K}$$

$$\text{Load} \rightarrow V_A \rightarrow R_B \rightarrow \text{Demand} = 0$$

$$\text{Fanout, } N = \frac{\text{Supply}}{\text{Demand}} = \infty$$



Case-II

$$V_o = \text{low} = 0.2V$$

$$I_{C_{\max}} = I_{C'} + N I_L \Rightarrow 12 = 2.18 + N \times 0.82$$

$$N = \text{Floor}(11.97)$$

$$I_{C'} \rightarrow \text{No load collector current} = 11$$

$$I_C = I_{C_{\max}} \rightarrow \text{Collector current limit for sat.}$$

$$I_L \rightarrow \text{Load current.}$$

$$I_{C'} = \frac{5-0.2}{2.2K} = 2.18 \text{ mA}$$

$$I_L = I_{OA} = \frac{5-0.2}{5K} = 0.82 \text{ mA}$$

$$\beta_{\text{forced}} = \beta_F = \frac{I_{C_{\max}}}{I_B} \Rightarrow 30 = \frac{I_{C_{\max}}}{0.4}$$

$$\therefore \text{Maximum Fanout, } N = \min(\infty, 11) \quad \boxed{I_{C_{\max}} = 12 \text{ mA}}$$

$$= 11$$

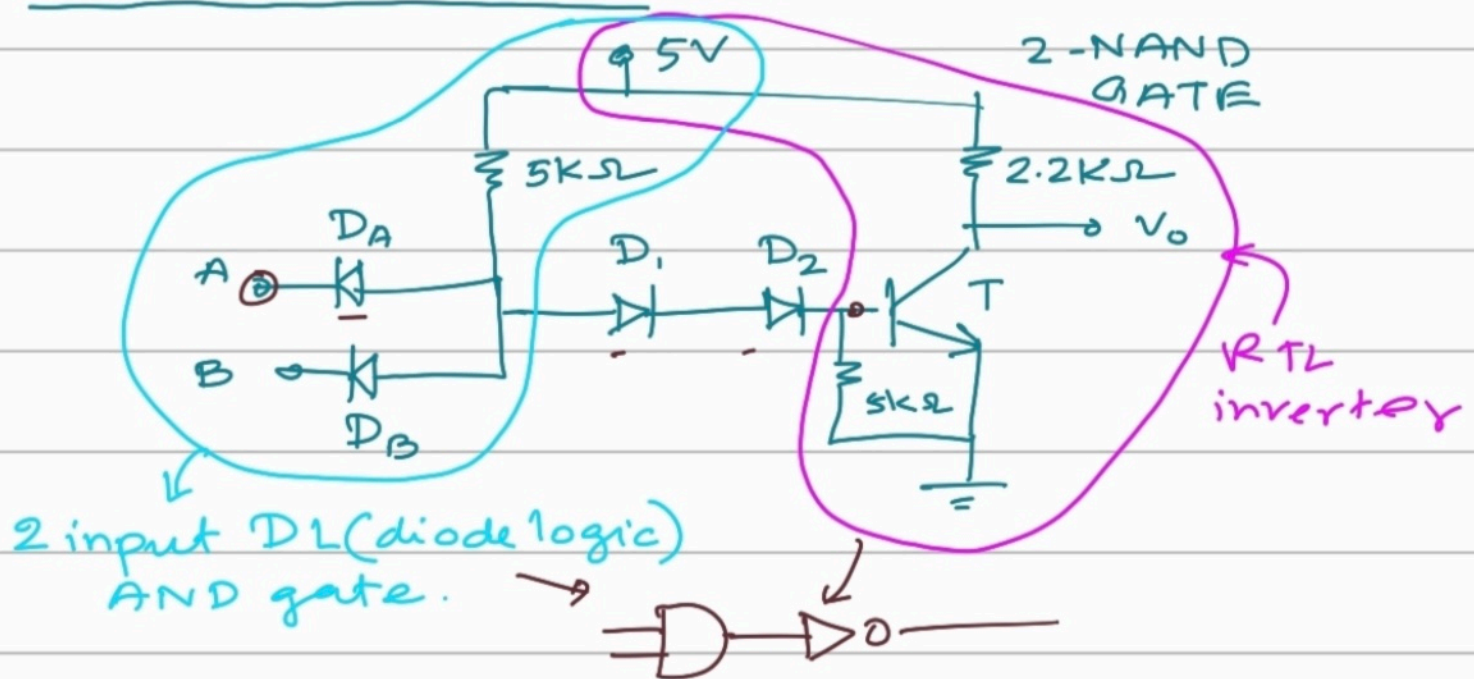
maximum fanout = ∞ = any answer is valid.

DTL (Diode Transistor Logic)

* Bipolar $\rightarrow$ Saturated
(Switching transistor will operate in saturation and cutoff mode).

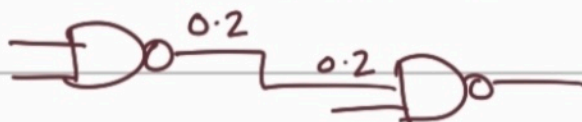
* Input terminals connects through diodes to the switching transistor's base terminal.

DTL NAND GATE:



Basic operation:

Calculate the currents and voltages of the following DTL gate and verify that it works like a NAND gate. Assume that input stages might be connected to the output stage of other DTL NAND gate.
 $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.2V$. $\beta_F = 30$.

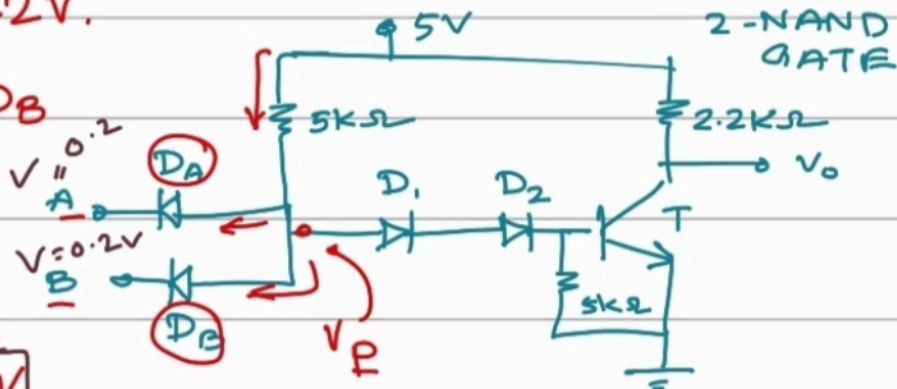
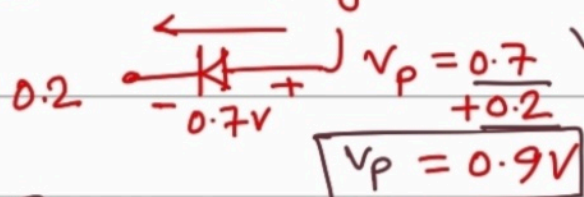


Ans:

Case ① Both inputs are low.

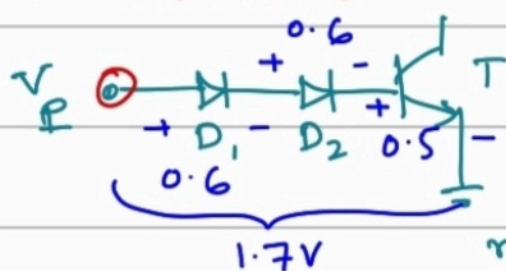
Because inputs are connected to some other output stage, the output voltage = input voltage = $0.2V$. $V_A = V_B = 0.2V$.

Assume, D_A and D_B are conducting.



Since we are assuming diodes are conducting the anode voltage V_P of D_A and D_B must be $0.7V$ higher than the cathode voltage. Thus $V_P = 0.9V$.

Necessary condition for transistor T to turn on:



$$V_{BE}(\text{cut in voltage}) = 0.5V$$

$$V_x(\text{cut in voltage of diode}) = 0.6V$$

In order for T to turn on we need to have $V_P \geq 1.7V$.

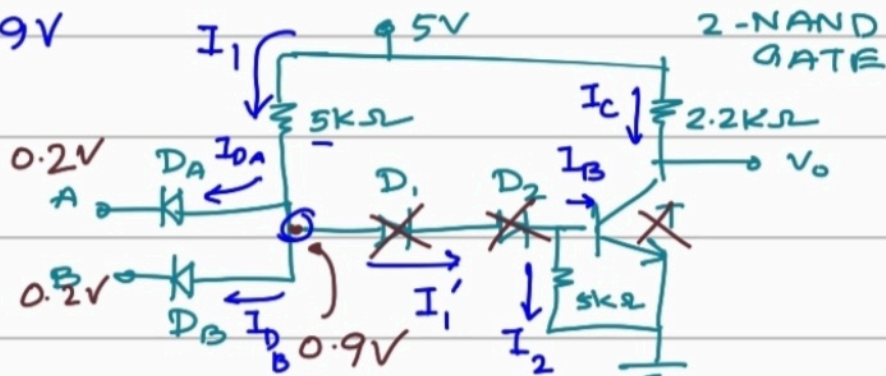
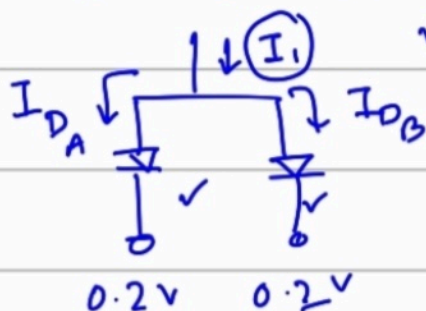
According to our assumption, V_P will not turn on D_1 , D_2 and T and current will flow as we expected.

$$V_A = V_B = 0.2V, V_P = 0.9V$$

$$I_1 = \frac{(5 - 0.9)}{5k} = 0.82mA$$

$$I_C = I_B = I_2 = I_1' = 0$$

$$V_0 = 5V$$



$I_{DA} = I_{DB}$ because both branches are identical.

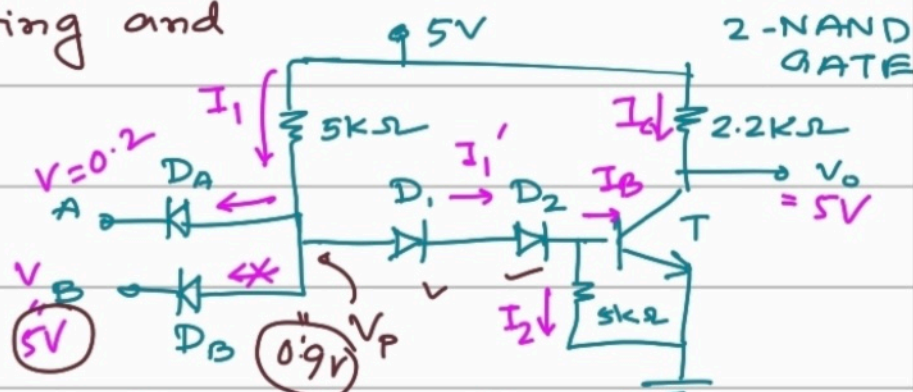
$$I_{DA} = I_1/2 = 0.41mA$$

Case ② $V_A = 0.2V, V_B = 5V$ | ($V_A = 5V, V_B = 0.2V$)
Similar

Assume. D_A is conducting and D_B is off.

Therefore, $V_P = 0.9V$.

So D_1, D_2, T must be turned off.



The anode of D_B has smaller voltage than the cathode. So our assumption is right.

$$\# I_C = I_B = I_1' = I_2 = 0, V_O = 5V. I_1 = \frac{5 - 0.9}{5k\Omega} = 0.82 \text{ mA}$$

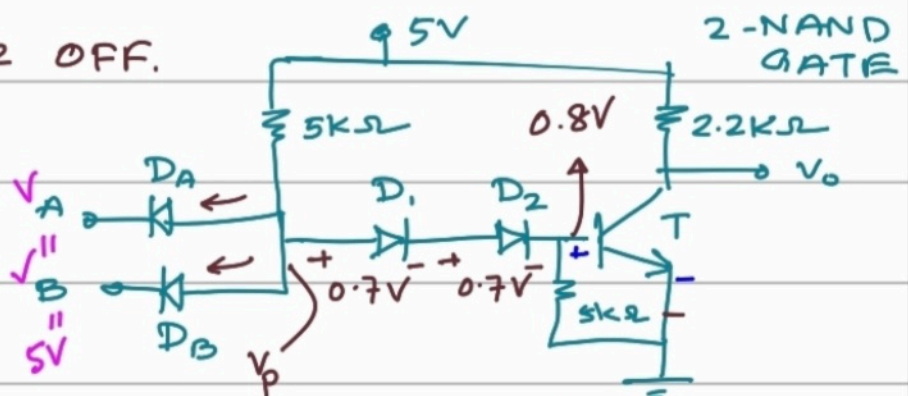
$$I_{D_A} = I_1 = 0.82 \text{ mA}, I_{D_B} = 0 \text{ mA}.$$

Case ③ $V_A = V_B = 5V$.

Assume, D_A and D_B are OFF.

$$I_{D_A} = I_{D_B} = 0 \text{ mA}.$$

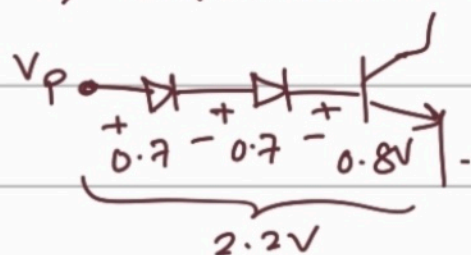
Therefore current must flow through D_1, D_2 and T .



Because it is a DTL circuit, the switching transistor must operate in saturation when it is turned on.

$$V_{BE(sat)} = 0.8V, V_{CE(sat)} = 0.2V$$

⇒ Conduction voltage of D_1 and D_2 diodes are $0.7V$.



According to our assumption

$$V_P \text{ must be } 2.2V.$$

The cathode voltages of D_A and D_B is higher than the anode voltage $V_P = 2.2V$

This justifies our assumption.

Current-voltage Calculation:

$$V_P = 2.2V$$

$$I_1 = \left(\frac{5 - 2.2}{5k} \right) = 0.56mA$$

$$I_{D_A} = I_{D_B} = 0mA$$

$$I_1' = I_1 = 0.56mA$$

$$I_2 = \frac{0.8 - 0}{5k} = 0.16mA$$

$$I_B = I_1' - I_2 = 0.56 - 0.16 = 0.4mA$$

$$I_C = \frac{5 - 0.2}{2.2k} = 2.182mA$$

Chart:

V_A	V_B	V_O
0.2	0.2	5V
0.2	5V	5V
5V	0.2	5V
5	5	0.2

From analyzing the cases we can clearly conclude that this circuit is a NAND gate.

β_{min} for this circuit:

$$\beta_{forced} = \frac{I_C}{I_B} = \frac{2.182}{0.4} = 5.45$$

We need to choose $\beta_F \geq 5.45$ for transistor T in order to prevent it from malfunctioning.

$$\beta_{min} = 5.45$$

Lecture 8: DTL NOISE MARGIN, FANOUT, Modified DTL, HTL

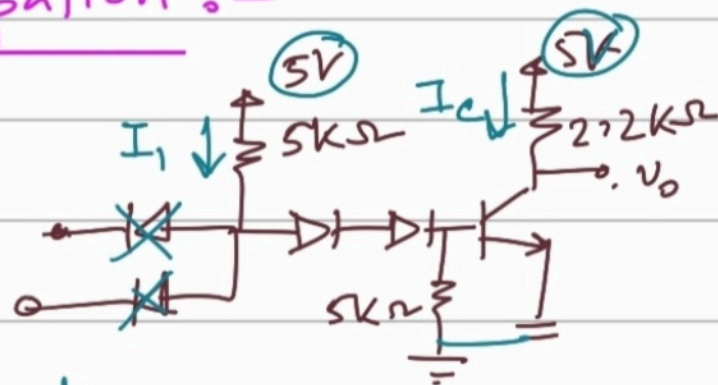
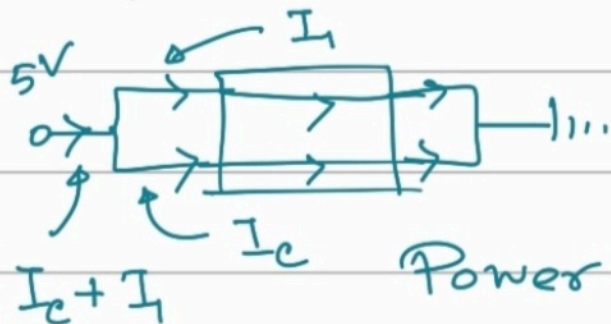
Noise Margin: Different calculation from RTL circuit.

V_{NH} (noise margin):

Q: If all inputs are high, then what is the magnitude of noise voltage which will cause the gate to malfunction?

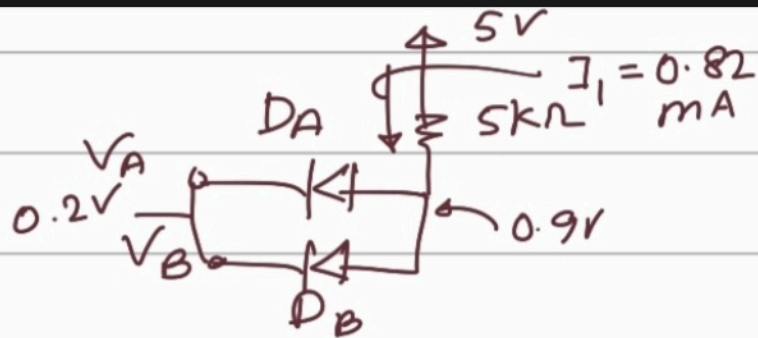
Maximum Power Dissipation :-

* Case ③ $V_A = V_B = 5V$



$$\begin{aligned} \text{Power dissipation} &= (5 - 0) \times (2.182 + 0.56) \\ &= \boxed{13.71 \text{ mW}} \end{aligned}$$

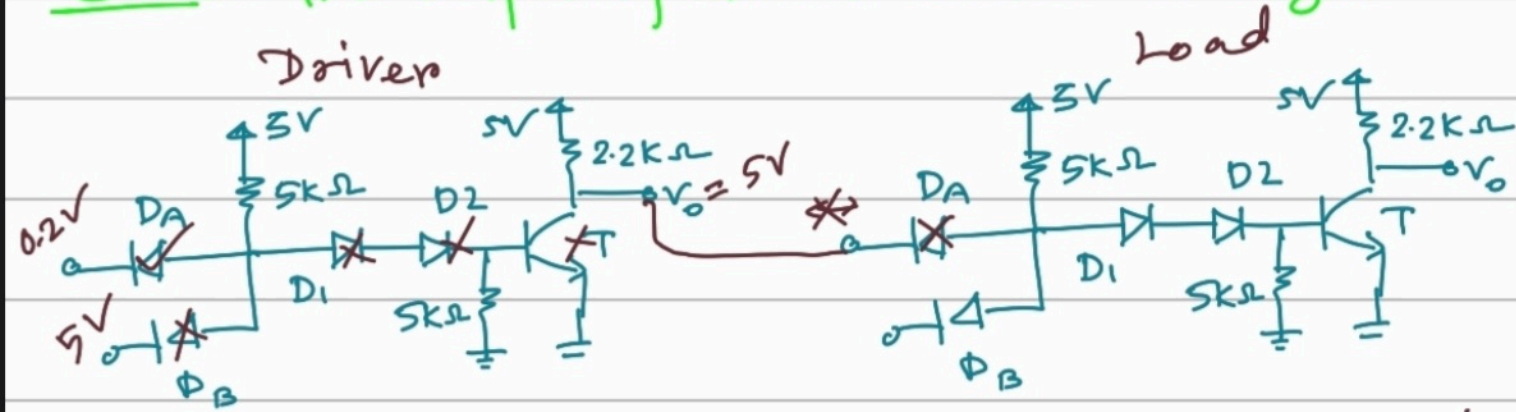
Case ① $V_A = V_B = 0.2$



$$\begin{aligned} \text{Power dissipation, } &(5 - 0.2) \times 0.82 \\ &= \boxed{3.936 \text{ mW}} \end{aligned}$$

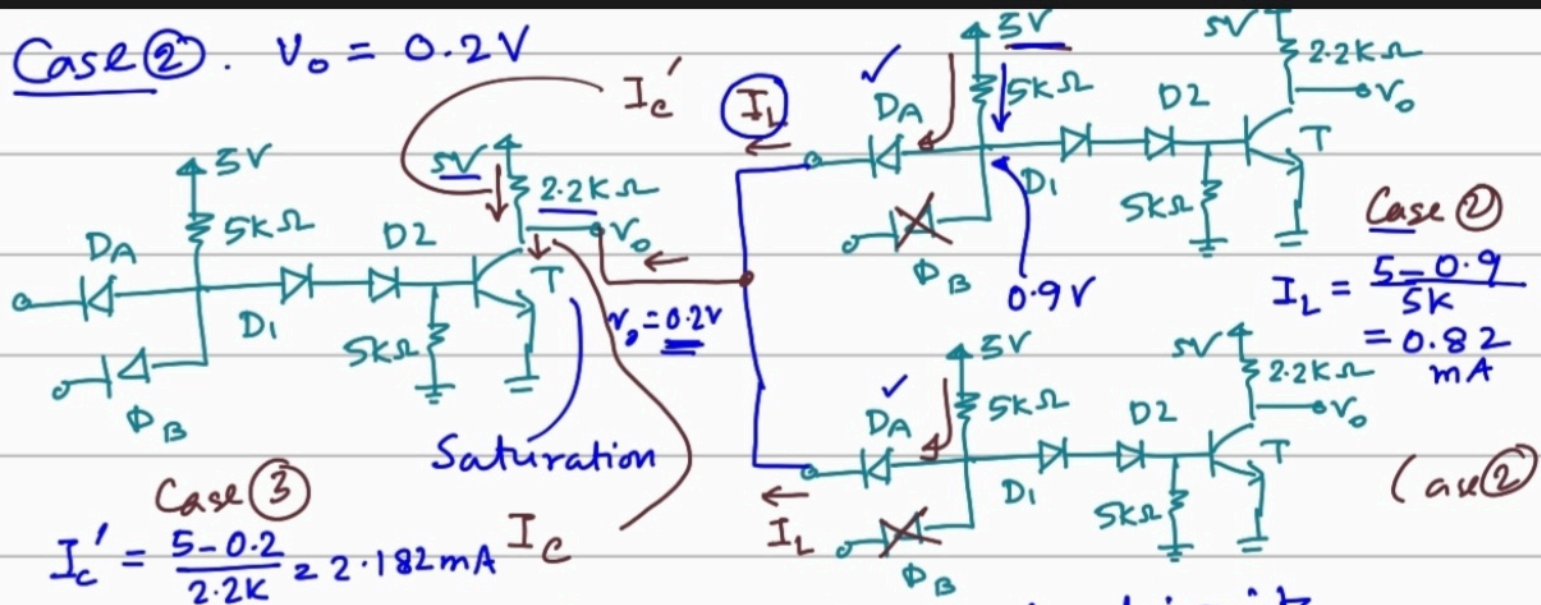
Maximum fanout Calculation :-

Case ①. The output of driver circuit is high.



The demand current is zero, and supply is zero, also. Thus we can connect as much as load circuits we want. $\text{max Fanout} = \infty$.

Case ②. $V_o = 0.2V$



We want to consider worst case scenario, thus we will consider D_B of load circuits are off.

I_c = total collector current, I_c' = no. load collector current.
 I_L = standard load (current), N = fanout

$$I_c = I_c' + N I_L = 2.182 + N \times 0.82$$

transistor

Constraint: When driver circuit is operating in saturation mode, the collector current can not exceed a certain value. Otherwise, the driver transistor T will change its operating mode saturation to forward active.

Marginal situation will be achieved if

$\beta_{\text{forced}} \approx \beta_F$ for driver transistor T .

$$\beta_{\text{forced}} \approx \frac{I_c}{I_B} \Rightarrow I_{c,\text{max}} = \beta_F \times I_B = 30 \times 0.4 = 1.2 \text{ mA}$$

← Case ③

Maximum fanout is N' . Then the main equation becomes, $I_{c,\text{max}} = I_c' + N' I_L$

$$1.2 = 2.182 + N' \times 0.82$$

$$\Rightarrow N' = 11.96$$

We can not choose maximum fanout $N' = 12$

because that will exceed the value of permitted collector current.

Therefore, maximum fanout for this case (N')

$$\text{Finally maximum fanout} = \min(11, \infty) = 11$$